ECE 447

Lesson 38
Error Correction: Parity, CRC, and Hamming Codes



UNITED STATES
AIR FORCE
ACADEMY

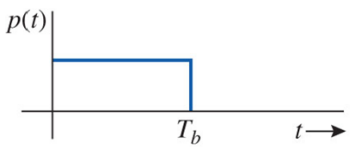
Schedule and Admin

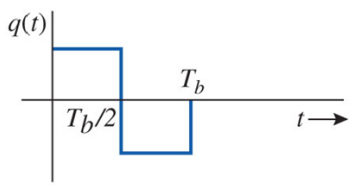
- Schedule.
  - Lesson 38 - Error correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
  - **Final Exam** – 0730-1120, Saturday, 12 Dec
- Admin
  - **HW9**. Due Lesson 40 (Mon, 7 Dec) to Gradescope.
  - **Lab 8**. Assigned Lesson 39 (Thu, 3 Dec); due Lesson 41 (Wed, 9 Dec) to Gradescope.
  - **Extra Credit** (optional). Due Lesson 41 (Wed, 9 Dec).

REVIEW

Matched Filters and Binary Signaling Performance

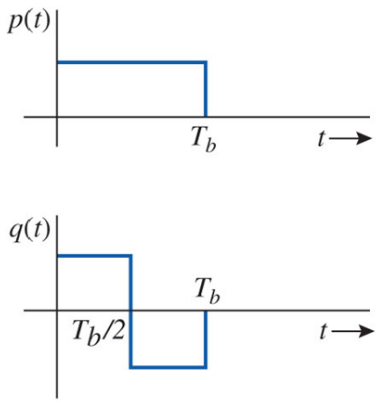
- Why?
- General binary case for real-valued pulses: $h(t) = p(T_b - t) - q(T_b - t)$
- $P_e = P_b = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$, where $E_{pq} = \int_0^{T_b} p(t)q(t)dt$ is the inner product or correlation of the two pulses
- Example problem: Find the matched filter impulse response $h(t)$ and calculate the total bit error probability P_e assuming equiprobable transmission of 0 or 1





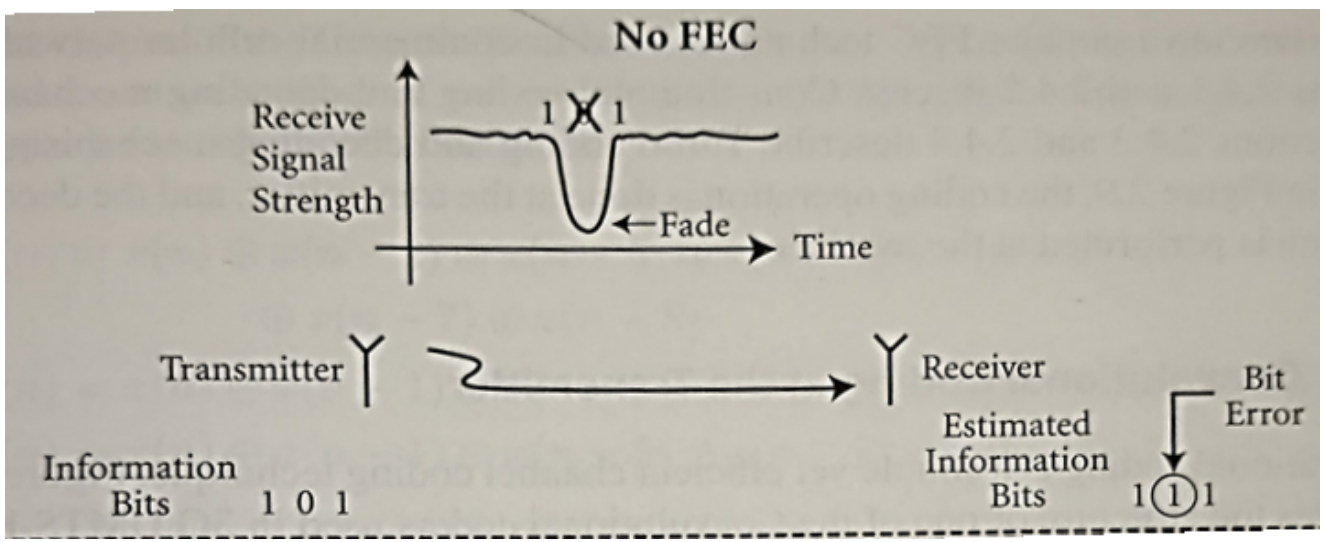
REVIEW

- Example problem: Find the BER for a system using **polar signaling** pulses $\pm 3p(t)$ with $T_b = 1\mu$s and $N_0 = \frac{25\mu\text{W}}{\text{Hz}}$.



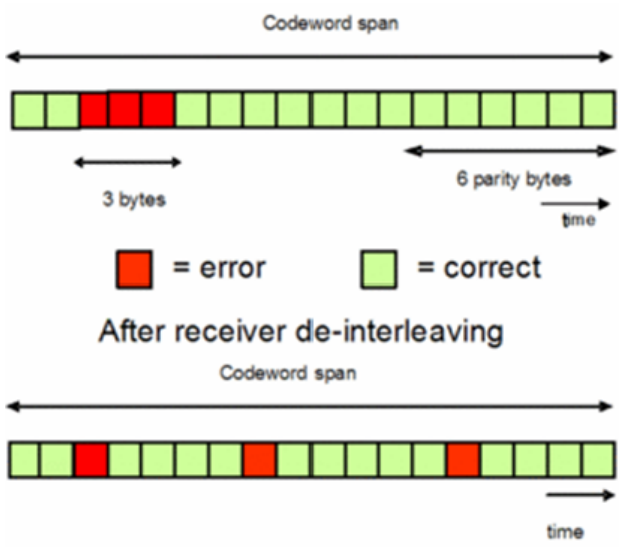
Error Correction

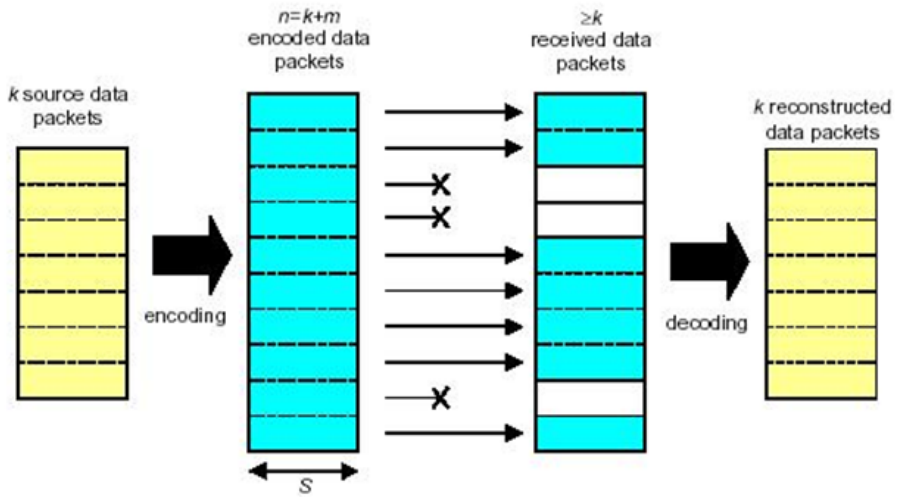
- Bit errors occur for a variety of reasons, but can correct many by pre-processing bits at the transmitter - **forward error correction (FEC)**
- "Error correction coding requires a strong mathematical background" - so true!
- Types of codes (in order of complexity): repetition, block, convolutional, Viterbi algorithm, turbo, low-density parity check (LDPC), polar



INTERLEAVING

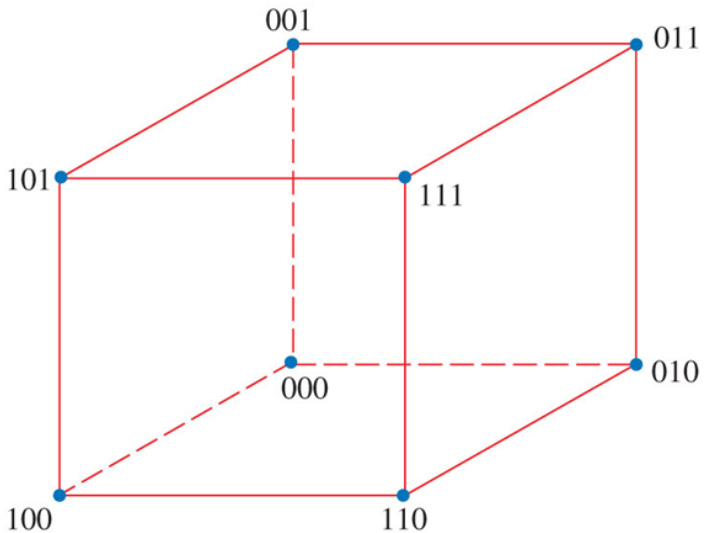
- First, though - errors often consecutive in time and/or frequency
- Interleaving spreads out code symbol errors at receiver - decoder receives few to zero consecutive errors
- Synchronization required!





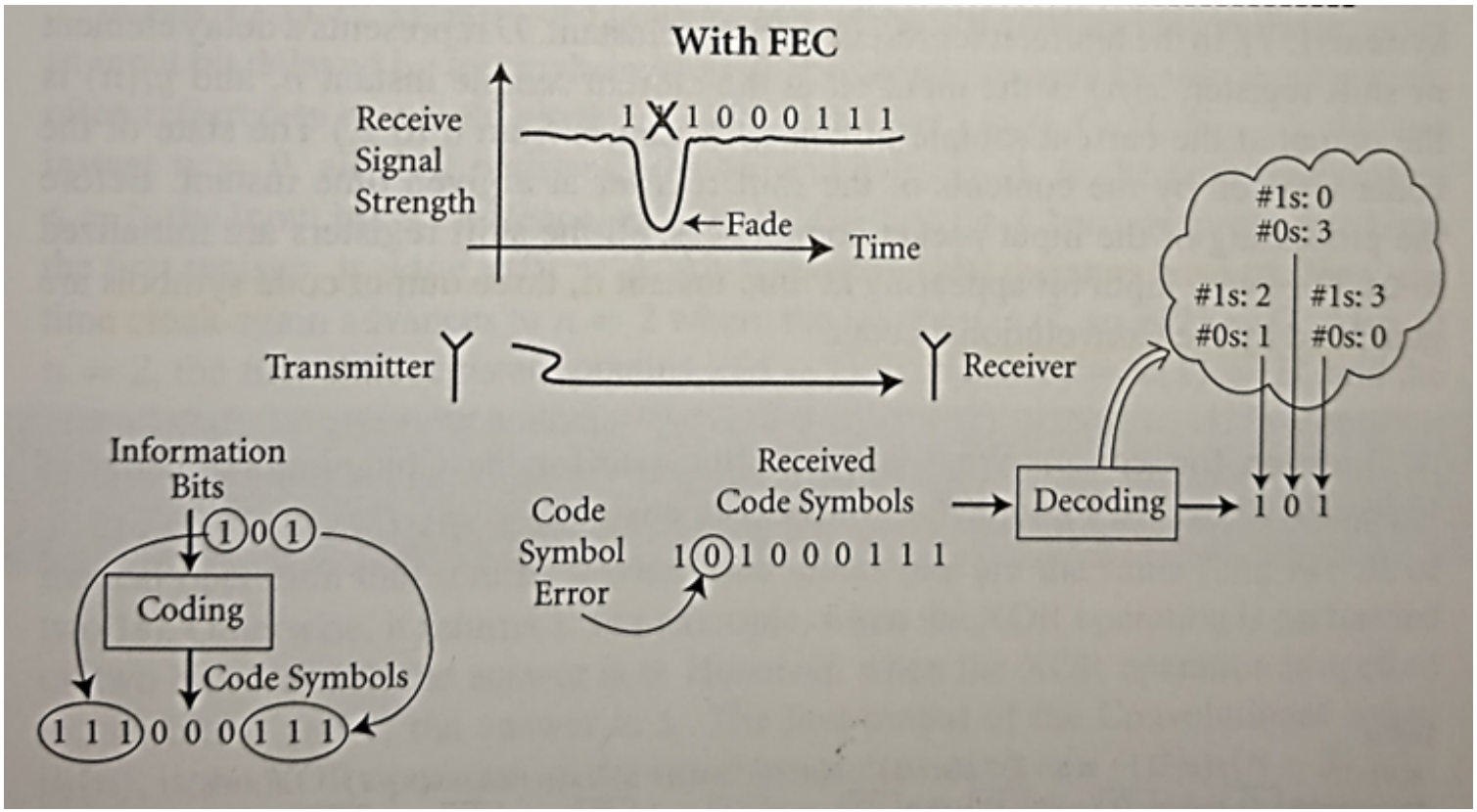
Error Correction Basics and Terminology

- **Code rate:** ratio of data bits k to total bits n - creates an (n, k) code
- d is bit vector **message** (data)
- c is bit vector **codeword**
- Number of redundant (aka check aka parity) bits is $m = n - k$
- **Hamming distance:** number of bits that are different between two bit vectors
- **Hamming space:** multi-dimensional cube; spheres drawn around possible codeword vertices - the radius spheres can have without overlapping equals the number of errors that can be corrected
- To *correct* t errors, min Hamming distance $d_{min} = 2t + 1$
- To *detect* t errors, min Hamming distance $d_{min} = t + 1$



Repetition Code

- Repeat each bit a certain number of times, then transmit
- To decode at receiver, use a simple majority rule



Hamming Code

- Type of linear block code
- $\boldsymbol{c} = \boldsymbol{dG}$, where $\boldsymbol{G}$ is called the **generator matrix**
- We transmit $\boldsymbol{c}$ and receive $\boldsymbol{r}$, which may have errors
- $\boldsymbol{s} = \boldsymbol{rH}^T$ is called the **syndrome**, where $\boldsymbol{H}$ is the **parity check matrix**
- The row in $\boldsymbol{H}^T$ that matches $\boldsymbol{s}$ is the bit of the codeword in error - flip that bit
- If $\boldsymbol{s} = \boldsymbol{0}$, then no errors
- **Coding gain:** reduction in req'd SNR to achieve specific BER

What do we "gain"?

- **Coding gain:** reduction in req'd SNR to achieve specific BER

